

SUNYU WANG

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EDUCATION

Carnegie Mellon University

Pittsburgh, Pennsylvania, USA

Ph.D. in Robotics

2021–2027(*expected*)

- Advisors: Prof. Nancy Pollard, Prof. Jean Oh

University of Illinois Urbana-Champaign

Urbana, Illinois, USA

M.S. in Mechanical Engineering

2019–2021

- Advisor: Prof. Joao Ramos
- Thesis: A whole-body human-machine interface for dynamic bilateral teleoperation of humanoid robots

B.S. in Mechanical Engineering, Minor in Electrical Engineering

2015–2019

- GPA: 3.92/4.00. Highest honors.

PUBLICATIONS

- [7] **Relative Contact Velocity-Controlled Hand-Object Mechanism for Dexterous Tool Manipulation.**
Sunyu Wang, Jean Oh, Nancy S. Pollard.
Under Review, 2026.
- [6] **Geodesic Tracing-Based Kinematic Integration of Rolling and Sliding Contact on Manifold Meshes for Dexterous in-Hand Manipulation.**
Sunyu Wang, Arjun S. Lakshmipathy, Jean Oh, Nancy S. Pollard.
IEEE-RAS 24th International Conference on Humanoid Robots (Humanoids), 2025.
- [5] **The Foundational Pose as a Selection Mechanism for the Design of Tool-Wielding Multi-Finger Robotic Hands.**
Sunyu Wang, Jean Oh, Nancy S. Pollard.
IEEE-RAS 24th International Conference on Humanoid Robots (Humanoids), 2025.
- [4] **Dynamic Locomotion Teleoperation of a Reduced Model of a Wheeled Humanoid Robot Using a Whole-Body Human-Machine Interface.**
Sunyu Wang, Joao Ramos.
IEEE Robotics and Automation Letters, 2022.
- [3] **A Comparison Between Joint Space and Task Space Mappings for Dynamic Teleoperation of an Anthropomorphic Robotic Arm in Reaction Tests.**
Sunyu Wang, Kevin Murphy, Dillan Kenney, Joao Ramos.
IEEE International Conference on Robotics and Automation (ICRA), 2021.
- [2] **A Spring-Aided Two-Dimensional Electromechanical Spine Architecture for Bio-Inspired Robots.**
Bonhyun Ku, Sunyu Wang, Arijit Banerjee.
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2019.
- [1] **A Distributed and Scalable Electromechanical Actuator for Bio-Inspired Robots.**
Bonhyun Ku, Yanpei Tian, Sunyu Wang, Elie Libbos, Shivang Agrawal, Arijit Banerjee.
IEEE International Electric Machines and Drives Conference (IEMDC), 2019.

AWARDS

- 3rd Place, IEEE Energy Conversion Congress and Exposition (ECCE) Student Project Demo Competition 2019
- University of Illinois Urbana-Champaign Mechanical Science and Engineering Departmental Distinguished Fellowship 2019–2020

TEACHING	16-711: Kinematics, Dynamic Systems, and Controls	
	Teaching Assistant and Guest Lecturer, Carnegie Mellon University	2024–2025
	CS 125: Introduction to Computer Science	
	Course Assistant, University of Illinois Urbana-Champaign	2016
MENTORING	Undergraduate	
	• Jinglun Ke, Carnegie Mellon University	2025
	• Lulu Zhao, Beijing Normal University → Cornell University Ph.D.	2024
	• Zoey Ballard, University of Houston	2024
	• Dillan Kenney, University of Illinois Urbana-Champaign → SpaceX	2020–2021